

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                                   |
|-----------------------------|---------------------------------------------------|
| <b>Product Name</b>         | <b><u>Charcoal, vegetable (Norit), powder</u></b> |
| <b>CAS No</b>               | 7440-44-0                                         |
| <b>Synonyms</b>             | Black Pearls; Charcoal Black; Graphite Nautural   |
| <b>Molecular Formula</b>    | C                                                 |
| <b>Molecular Weight</b>     | 12                                                |
| <b>Recommended Use</b>      | Laboratory chemicals.                             |
| <b>Uses advised against</b> | No Information available                          |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>A16079</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

**HSNO Approval Number**      **HSR002522**

### GHS Classification

#### Physical hazards

Self-heating substances/mixtures

Category 2

#### Health hazards

Based on available data, the classification criteria are not met

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements

None required

**Signal Word****Warning****Hazard Statements**

H252 - Self-heating in large quantities; may catch fire

**Precautionary Statements****Prevention**

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P314 - Get medical advice/attention if you feel unwell

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

## Section 3 - Composition and Information on Ingredients

| Component          | CAS No    | Weight % |
|--------------------|-----------|----------|
| Activated charcoal | 7440-44-0 | 100      |

## Section 4 - First Aid Measures

**Description of first aid measures**

|                                            |                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                              |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. If symptoms persist, call a physician.              |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Ingestion</b>                           | Drink plenty of water. Do NOT induce vomiting. Get medical attention if symptoms occur.                                 |
| <b>Self-Protection of the First Aider</b>  | No special precautions required.                                                                                        |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                    |
| <b>Most important symptoms and effects</b> | No information available.                                                                                               |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                  |

## Section 5 - Fire Fighting Measures

### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

### **Extinguishing media which must not be used for safety reasons**

No information available.

### **Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Carbon oxides.

### **Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### **Emergency procedures**

Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin and eyes. Use personal protective equipment as required.

#### **Environmental Precautions**

Should not be released into the environment.

#### **Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Provide adequate ventilation.

#### **Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

#### **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### **Advice on safe handling**

Avoid contact with skin and eyes. Avoid dust formation. Ensure adequate ventilation. Avoid breathing dust/fume/gas/mist/vapors/spray.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### Conditions for Safe Storage, Including any Incompatibilities

#### **Storage Conditions**

Keep containers tightly closed in a cool, well-ventilated place.

#### **Incompatible Materials**

Incompatible Materials. Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

The product does not contain any hazardous materials with occupational exposure limits established.

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

### Appropriate engineering controls

#### Engineering Measures

None under normal use conditions.

### Individual protection measures, such as personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber, Neoprene, Natural rubber, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Particle filter (or AUS/NZ equivalent)

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                     |                   |
|---------------------|-------------------|
| Physical State      | Solid             |
| Appearance          | Black             |
| Odor                | Odorless          |
| Odor Threshold      | No data available |
| pH                  | No data available |
| Melting Point/Range | 3652 °C           |
| Softening Point     | No data available |

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Boiling Point/Range</b>                     | No data available        |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No data available        | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 452 °C                   |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | Not applicable           | Solid                                    |
| <b>Water Solubility</b>                        | Insoluble                |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Vapor Pressure</b>                          | 1 mmHg @ 3586 °C         |                                          |
| <b>Density / Specific Gravity</b>              | 1.8 - 2.1                |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Vapor Density</b>                           | Not applicable           | Solid                                    |
| <b>Particle characteristics</b>                | No data available        |                                          |
| <b><u>Other information</u></b>                |                          |                                          |
| <b>Molecular Formula</b>                       | C                        |                                          |
| <b>Molecular Weight</b>                        | 12                       |                                          |
| <b>Evaporation Rate</b>                        | Not applicable - Solid   |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                                           |
|-----------------------------------------|-----------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                |
| <b>Stability</b>                        | Stable under normal conditions.                           |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                  |
| <b>Sensitivity to Static Discharge</b>  | No information available                                  |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                  |
| <b>Hazardous Reactions</b>              | No information available.                                 |
| <b>Conditions to Avoid</b>              | Avoid dust formation, Protect from moisture, Excess heat. |
| <b>Incompatible Materials</b>           | Incompatible Materials, Strong oxidizing agents.          |
| <b>Hazardous Decomposition Products</b> | Carbon oxides.                                            |

## Section 11 - Toxicological Information

### Acute Effects

### Information on likely routes of exposure

#### Product Information

|                   |                                                |
|-------------------|------------------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure.             |
| <b>Eyes</b>       | Not an expected route of exposure.             |
| <b>Skin</b>       | No known effect based on information supplied. |
| <b>Ingestion</b>  | No known effect based on information supplied. |

### Numerical measures of toxicity

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| <b>Oral</b>                | Based on available data, the classification criteria are not met |

**Dermal  
Inhalation**

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

| Component          | LD50 Oral                  | LD50 Dermal | LC50 Inhalation |
|--------------------|----------------------------|-------------|-----------------|
| Activated charcoal | LD50 > 10000 mg/kg ( Rat ) |             |                 |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

(d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity;

Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Based on available data, the classification criteria are not met

(i) STOT-repeated exposure;

Target Organs

Based on available data, the classification criteria are not met  
None known.

(j) aspiration hazard;

Not applicable  
Solid

### Symptoms / effects, both acute and delayed

No information available.

## Section 12 - Ecological Information

### Ecotoxicity

Aquatic ecotoxicity

.

Terrestrial ecotoxicity

There is no data for this product

Persistence and Degradability

No information available

Bioaccumulative Potential

No information available

Mobility

No information available.

### Other adverse effects

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations .

## Section 14 - Transport Information

| Component                               | Hazchem Code |
|-----------------------------------------|--------------|
| Activated charcoal<br>7440-44-0 ( 100 ) | 1Y           |

**NZS 5433:2020**

Not regulated

**IATA**

Not regulated

**IMDG/IMO**

Not regulated

**Environmental hazards**

No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**

Not applicable, packaged goods

**Special Precautions**

No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information**

None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                      |           |
|----------------------|-----------|
| HSNO Approval Number | HSR002522 |
|----------------------|-----------|

**National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Authorisation/Restrictions according to EU REACH

| Component          | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Activated charcoal | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component          | CAS No    | NZIoC | AICS | EINECS | ELINCS | NLP | KECL     | IECSC | TCSI |
|--------------------|-----------|-------|------|--------|--------|-----|----------|-------|------|
| Activated charcoal | 7440-44-0 | X     | X    | -      | -      | -   | KE-04671 | X     | X    |

| Component          | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|--------------------|-----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Activated charcoal | 7440-44-0 | X    | ACTIVE                                        | X   | -    | X     | -    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

#### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration



**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

#### Revision Date

22-Mar-2023

#### Revision Summary

Initial Release

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet